Supporting Information 2

ADDITIONAL RESULTS ON REACTION EQUILIBRIA

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1. What is this document about?

This document summarizes representative results obtained with *Notebook 2* when applied to elementary gas-phase reactions beyond the $N_2O_4 \rightleftharpoons 2NO_2$ case study. The goal is to illustrate the generality of the notebook workflow and to provide reference outputs for comparison and teaching use. Specifically, we analyze the following equilibria:

- $N_2 + 3\,H_2 \rightleftharpoons 2\,NH_3$
- $O_2 + 2\,H_2 \rightleftharpoons 2\,H_2O$

2. The $N_2 + 3H_2 \leftrightharpoons 2NH_3$ equilibrium

When executing Notebook 2, this reaction is defined by the following string:

- `N2 + 3 H2 -> 2 NH3`

The rest of data to be inserted in the Notebook can be found in Table S1.

Table S1. Input data for the $N_2 + 3H_2 \rightleftharpoons 2NH_3$ reaction. $\Delta_f H^\circ$ is reported in kJ·mol^{-1}, and S_m° and $C_{p,m}^\circ$ in J·mol^{-1}·K^{-1}. These three values correspond to 298 K.[1]

	$\Delta_f H^\circ$	S_m°	$C_{p,m}^\circ$	PubChem CID	Molecular charge	Number of unpaired electrons	Symmetry number
N_2	0.00	191.61	29.125	947	0	0	2
H_2	0.00	130.684	28.824	783	0	0	2
NH_3	-46.11	192.45	35.06	222	0	0	3

For a initial number of moles equivalent to the stequimetric coefficients for reactants (i.e. 1 mol for N_2 and 3 for H_2, none for NH_3) the variation of $G(\xi)$ and of $A(\xi)$ with the extent of reaction ξ can be found at Figs. S1 and S2, respectively.

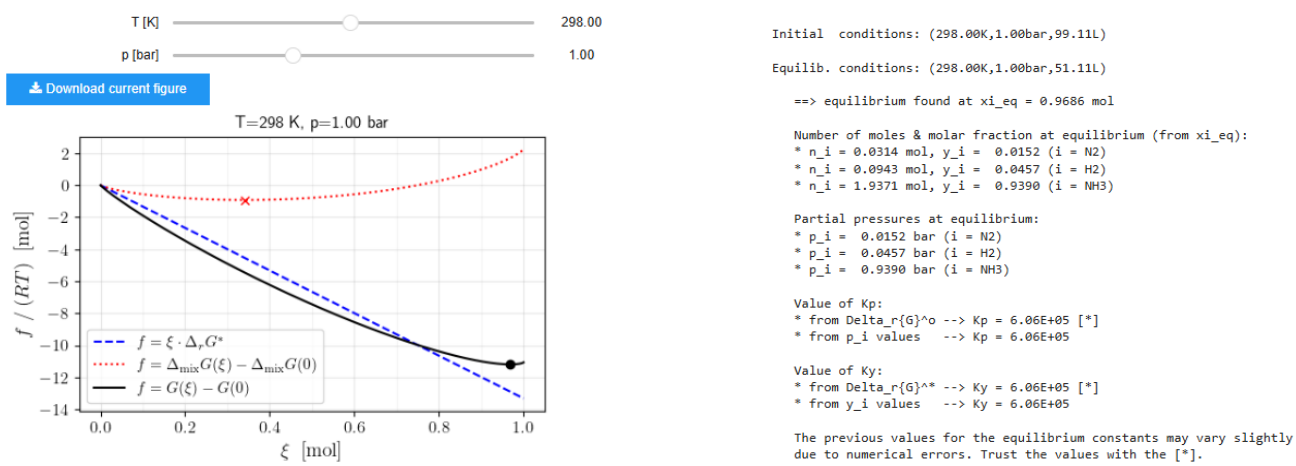


Figure S1. Screenshots showing equilibrium results at constant p and T. Sliders above the graph can be moved and the graph updates accordingly.

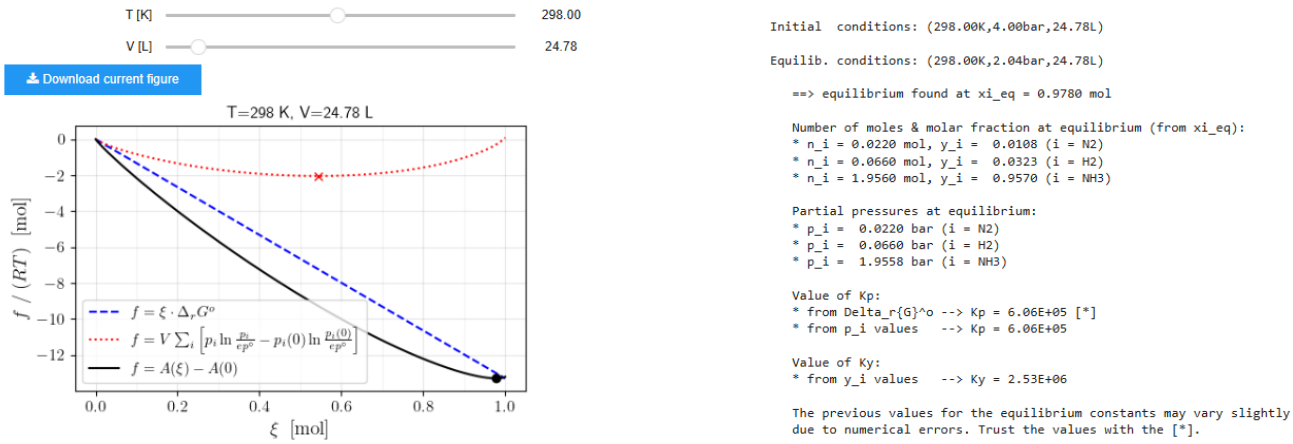


Figure S2. Screenshots showing equilibrium results at constant V and T. Sliders above the graph can be moved and the graph updates accordingly.

Entering in the Statistical-Mechanical Perspective, the user has to introduce the PubChem CID to retrieve the initial geometry of the reactants and product. These numbers can be found in Table S1. An example of the results can be found in Fig. S3

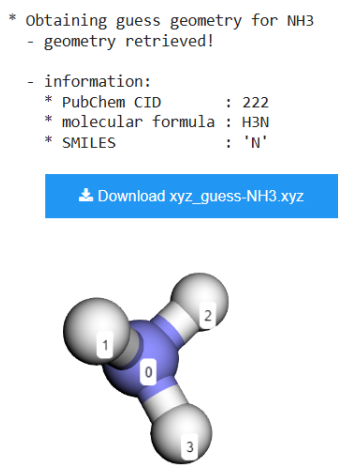


Figure S3. Screenshots showing the retrieve structure for NH3. The view is interactive and the molecule can be visualized from different angles.

For this system, we recommend to carry out the calculations with the PBE0 level in combination with the 6-31G* basis set (this level of theory was selected based on benchmarking against experimental data).[1, 2]. The vibrational frequencies associated to the optimized geometries are shown in Table S2.

Table S2. Comparison between experimental[a] and theoretical vibrational frequencies (in cm^{-1}) for H_2, N_2 and NH_3

	H_2	N_2	NH_3
Exp[a]	4401	2359	950, 1627 (2), 3337, 3444 (2)
PBE0/6-31G*	4451	2498	1131, 1733 (2), 3486, 3627 (2)

[a]Experimental data taken from NIST Chemistry WebBook.[2]

Once the symmetry numbers are corrected according to Table S1, the themodynamic data shown in Table S3 is obtained.

Table S3. Comparison between experimental[a] and theoretical determinations of $\Delta_r H°$, $\Delta_r S°$, $\Delta_r G°$ and equilibrium constants for the reaction $N_2 + 3H_2 \rightleftharpoons 2NH_3$ at 298 K

	$\Delta_r H°/(\mathrm{kJ/mol})$	$\Delta_r S°/(\mathrm{J/mol/K})$[a]	$\Delta_r G°/(\mathrm{kJ/mol})$
Exp[a]	-92.22	-198.8	-32.99
PBE0/6-31G*	-90.99	-198.0	-32.00

[a]Experimental data taken from Atkins' Physical Chemistry.[1]

Moreover, the temperature dependence of $\Delta_r G°$ shown in Fig. S4 is obtained. Notice that perfect agreement is obtained when compared with the results obtained using the classical thermodynamics expression if $\Delta_r C_{p,m}^{\circ}$ is set to -5.71·R.

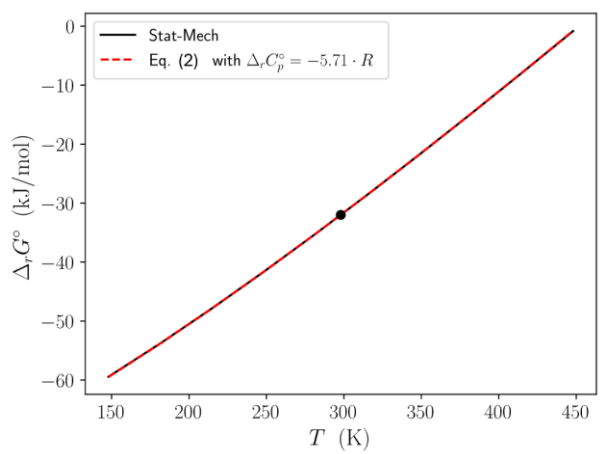
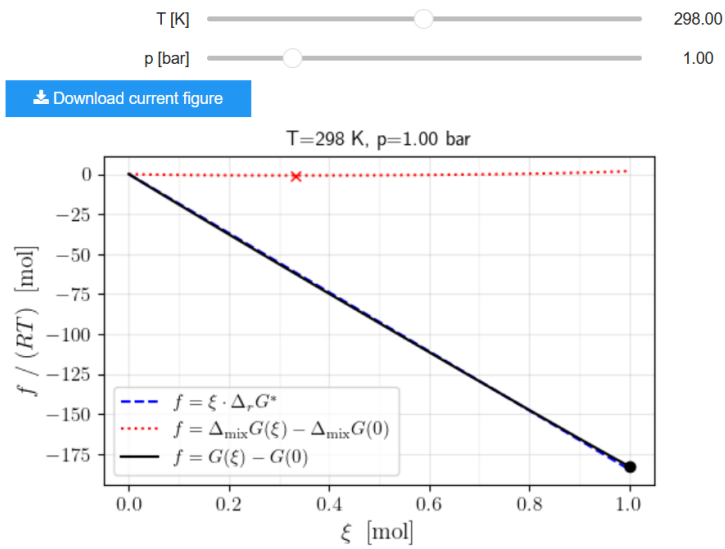


Figure S4. Standard Gibbs energy of reaction as a function of T, obtained from statistical mechanics (black curve) and by applying Eq. (2) of the manuscript (red dashed curve). The values of $\Delta_r H°$, and $\Delta_r S°$ at 298 K used in this equation were obtained at the PBE0/6-31G* level of theory. The value of $\Delta_r C_p^\circ = -5.71 \cdot R$ has been obtained through fitting to the statistical-mechanical result.

3. The $2H_2 + O_2 \leftrightharpoons 2H_2O$ equilibrium

Figures S4 presents the results from Notebook 1 for this equilibrium at constant p and T. The sliders enable students to adjust the reaction conditions interactively.




```
Initial  conditions: (298.00K,1.00bar,74.33L)

Equilib. conditions: (298.00K,1.00bar,49.55L)

   ==> equilibrium found at xi_eq = 1.0000 mol

   Number of moles & molar fraction at equilibrium (from xi_eq):
   * n_i = 0.0000 mol, y_i =  0.0000 (i = H2)
   * n_i = 0.0000 mol, y_i =  0.0000 (i = O2)
   * n_i = 2.0000 mol, y_i =  1.0000 (i = H2O)

   Partial pressures at equilibrium:
   * p_i =  0.0000 bar (i = H2)
   * p_i =  0.0000 bar (i = O2)
   * p_i =  1.0000 bar (i = H2O)

   Value of Kp:
   * from Delta_r{G}^o --> Kp = 1.35E+80

   Value of Ky:
   * from Delta_r{G}^* --> Ky = 1.35E+80
```

Figure S4. Screenshot showing equilibrium results at constant p and T. Experimental thermodynamic parameters are taken from *Atkins' Physical Chemistry*.[1] The initial amounts of H_2 and O_2 are 2 and 1 mol, respectively.

Tables S3 and S4 report the standard enthalpy, entropy and Gibbs energy of reaction, as well as the vibrational frequencies, obtained with Notebook 2 at the M06-2X/aug-cc-pVDZ level of theory. This level of theory was selected based on benchmarking against experimental data.[1, 2] Note that, to properly perform the electronic-structure calculations for O_2, the correct open-shell electronic configuration must be specified (two unpaired electrons).

Table S3. Comparison between experimental[a] and theoretical determinations of $\Delta_r H°$, $\Delta_r S°$, $\Delta_r G°$ and equilibrium constants for the reaction $2H_2 + O_2 \rightleftharpoons 2H_2O$ at 298 K

	$\Delta_r H°/(\mathrm{kJ/mol})$	$\Delta_r S°/(\mathrm{J/mol/K})^a$	$\Delta_r G°/(\mathrm{kJ/mol})$
Exp[a]	-483.6	-88.64	-457.2
M06-2X/aug-cc-pVDZ	-477.7	-88.58	-451.3

[a]Data taken from Atkins' Physical Chemistry.[1]

Table S4. Comparison between experimental[a] and theoretical vibrational frequencies (in cm^{-1}) for H_2, O_2 and H_2O

	H_2	O_2	H_2O
Exp[a]	4401	1580	1595, 3657, 3756
M06-2X/aug-cc-pVDZ	4477	1771	1614, 3861, 3971

[a]Data taken from NIST Chemistry WebBook[2].

Figure S5 shows the statistical-mechanical results for the temperature dependence of the standard Gibbs energy of reaction, compared with the results obtained using the classical thermodynamics expression given by equation (2).

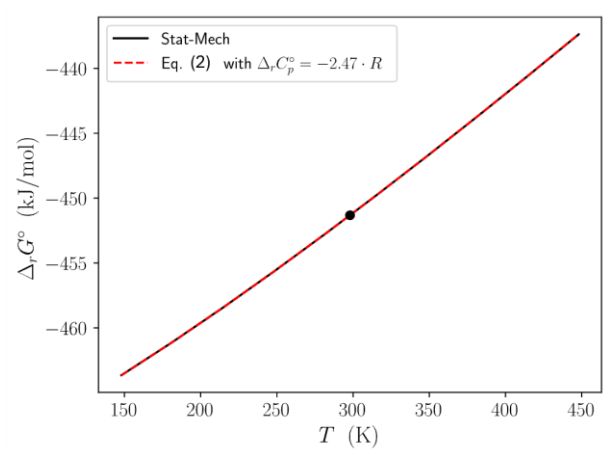


Figure S5. Standard Gibbs energy of reaction as a function of T, obtained from statistical mechanics (black curve) and by applying equation (2) (red dashed curve). The reference temperature is $T_{\mathrm{ref}} = 298$ K. The values of $\Delta_r H°(T_{\mathrm{ref}})$, and $\Delta_r S°(T_{\mathrm{ref}})$ used in equation (2) were obtained at the M06-2X/aug-cc-pVDZ level of theory. The value of $\Delta_r C_p^\circ = -2.47 \cdot R$ has been obtained through fitting to the statistical-mechanical result.

4. References

[1] Atkins, P. W.; De Paula, J.; Keeler, J. *Atkins' Physical Chemistry*; Oxford University Press, **2018**.

[2] Linstrom, P. J.; Mallard, W. G. *J. Chem. Eng. Data* **2001**, *46*, 1059-1063.